CCSA104 LAB
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Task 1: Explore the Default Policy

1. Policy Package in Use: The policy package currently in use is **Standard**.

2. Default Cleanup Rule Values

Field	Value
Name	Cleanup rule
Source	Any
Destination	Any
Services & Applications	Any
Action	Accept
Track	None
Install On	Policy Targets

3. Rulebase Column Configuration: The **Hits**, **Track**, and **Install On** columns were enabled by right-clicking a rulebase column header and selecting the required columns. One column was hidden and then restored successfully, confirming that column visibility can be customized within the rulebase view.

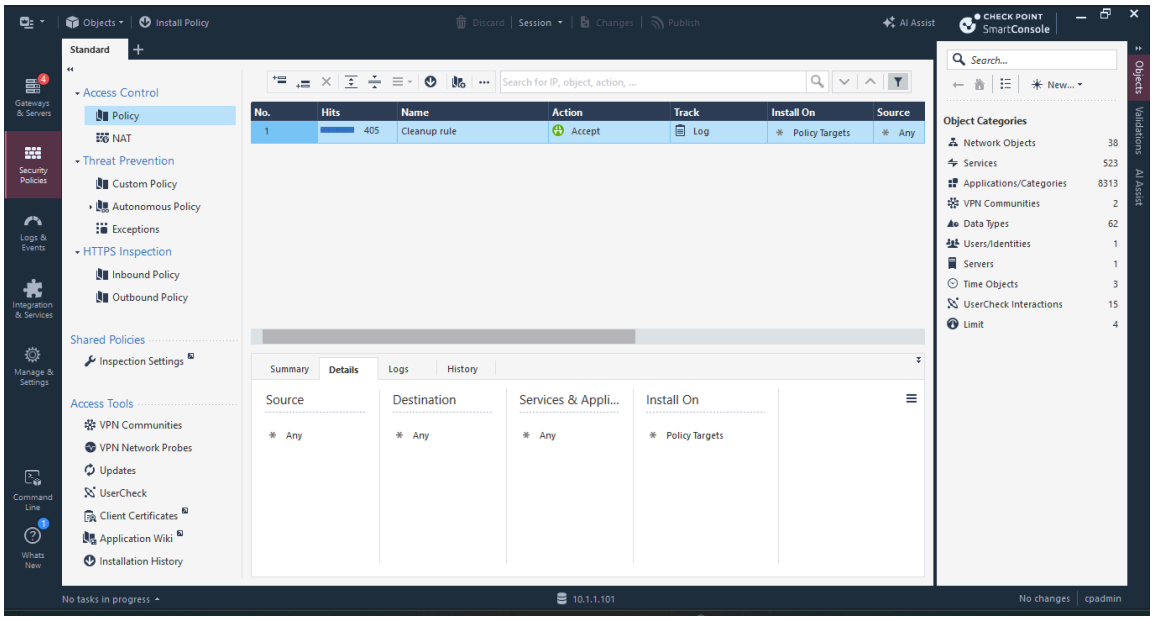
4. Global Properties and Policy Packages

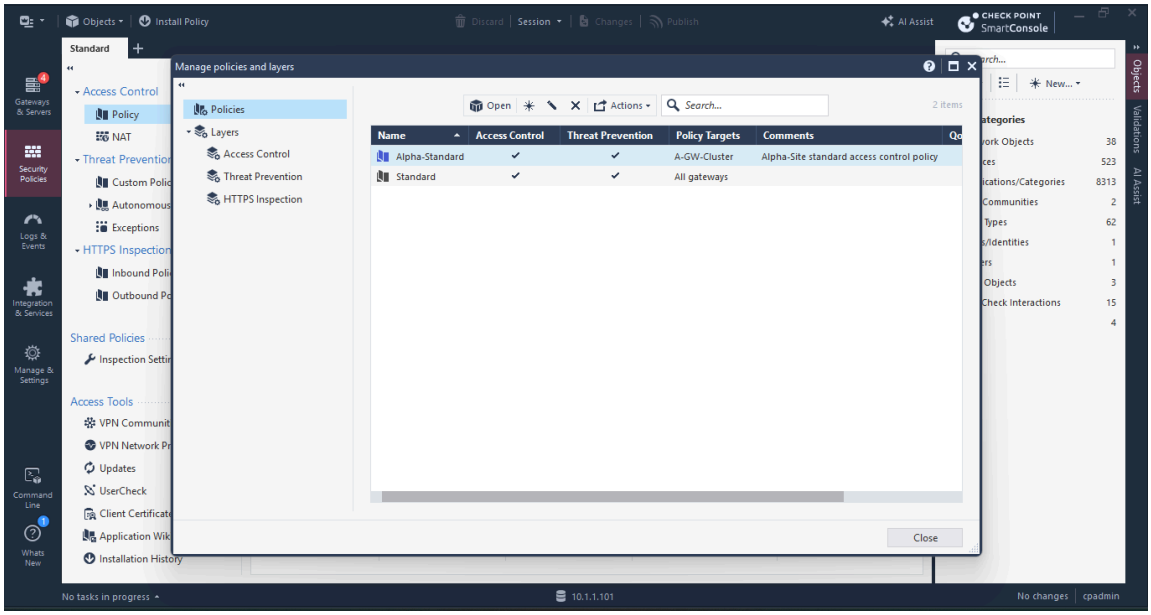
Global Properties Categories Observed:

1. Firewall
2. VPN
3. Remote Access

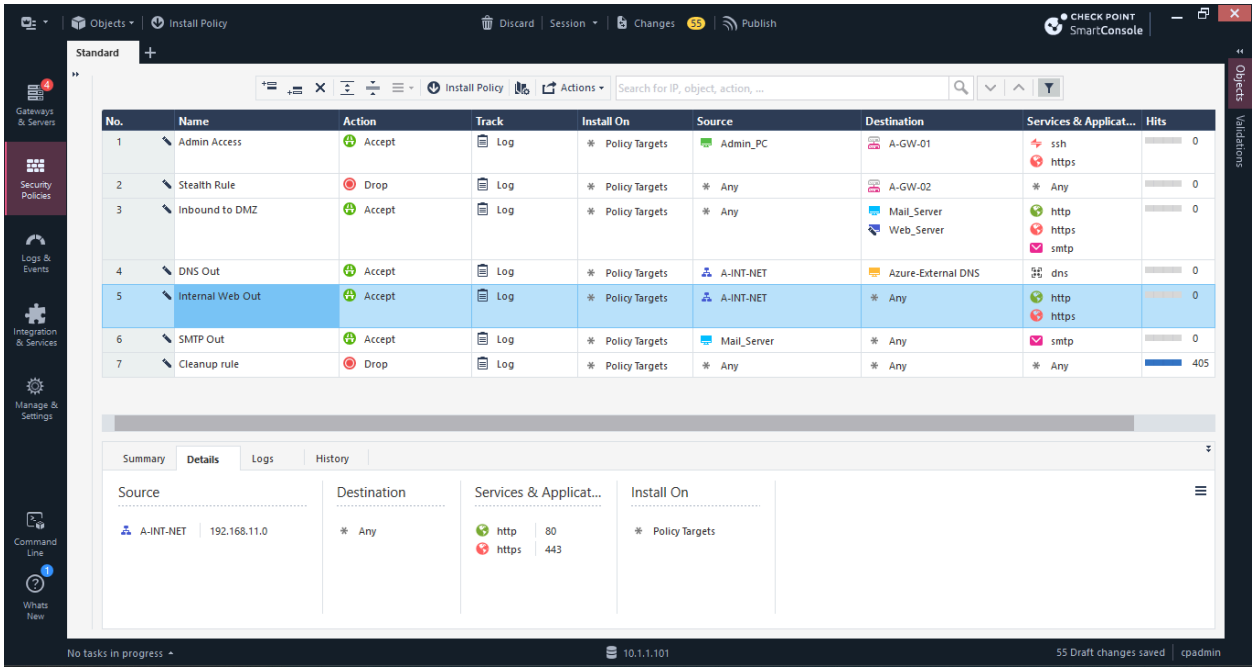
Policy Packages Present:

Policy Package	Access Control	Threat Prevention	Policy Targets
Standard	Enabled	Enabled	All gateways
Alpha-Standard	Enabled	Enabled	A-GW-Cluster

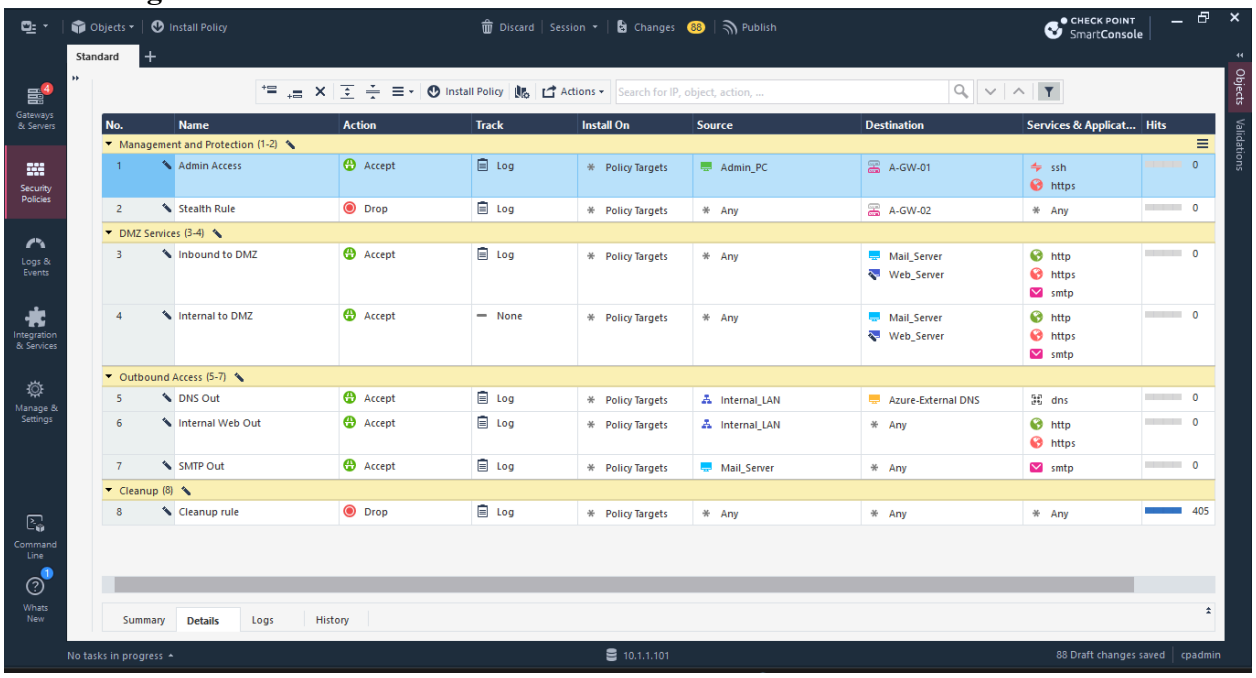

Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Standard
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
What's New
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Application Wiki
Installation History
Search for IP, object, action, ...
No.
Hits
Name
Action
Track
Install On
Source
1
405
Cleanup rule
Accept
Log
Policy Targets
Any
Summary
Details
Logs
History
Source
Destination
Services & Appli...
Install On
Any
Any
Any
Policy Targets
Search...
New...
Object Categories
Network Objects 38
Services 523
Applications/Categories 8313
VPN Communities 2
Data Types 62
Users/Identities 1
Servers 1
Time Objects 3
UserCheck Interactions 15
Limit 4
Objects
Validations
AI Assist
No tasks in progress
10.1.1.101
No changes
cpadmin

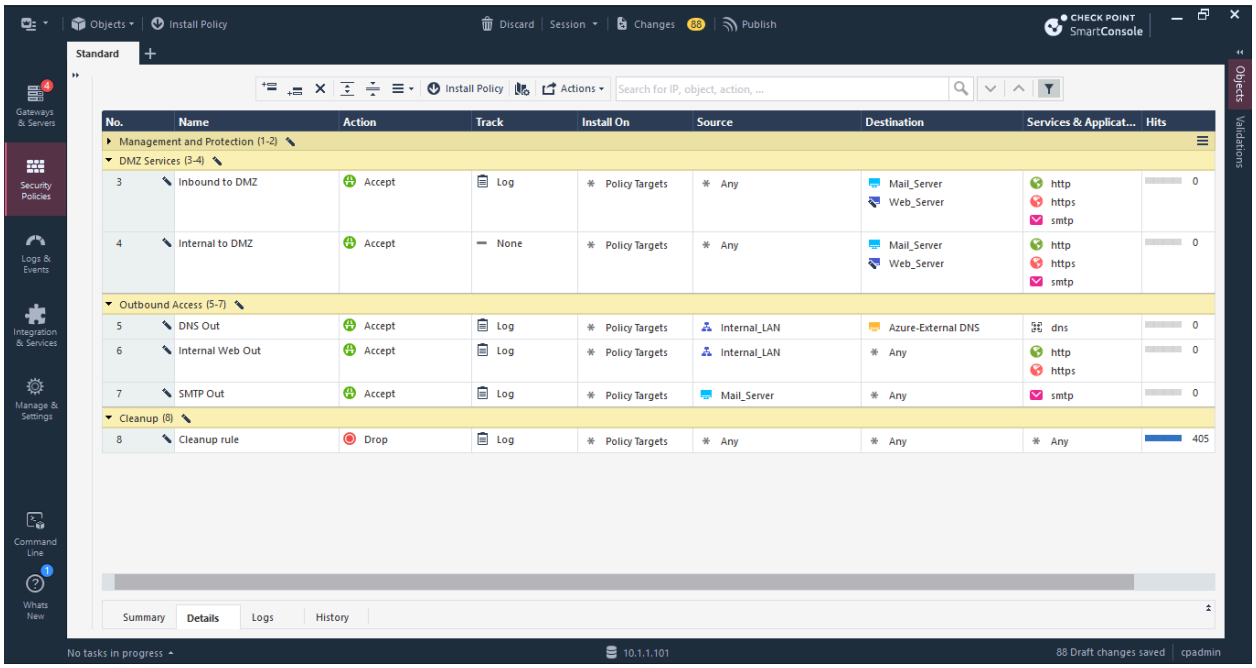

Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Standard
Manage policies and layers
Policies
Layers
Access Control
Threat Prevention
HTTPS Inspection
Open
Actions
Search...
2 items
Name
Access Control
Threat Prevention
Policy Targets
Comments
Alpha-Standard
A-GW-Cluster
Alpha-Site standard access control policy
Standard
All gateways
Close
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
What's New
Access Control
Policy
NAT
Threat Prevention
Exceptions
HTTPS Inspection
Shared Policies
Access Tools
Updates
UserCheck
Objects
Validations
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Task 2: Build the Access Control rulebase



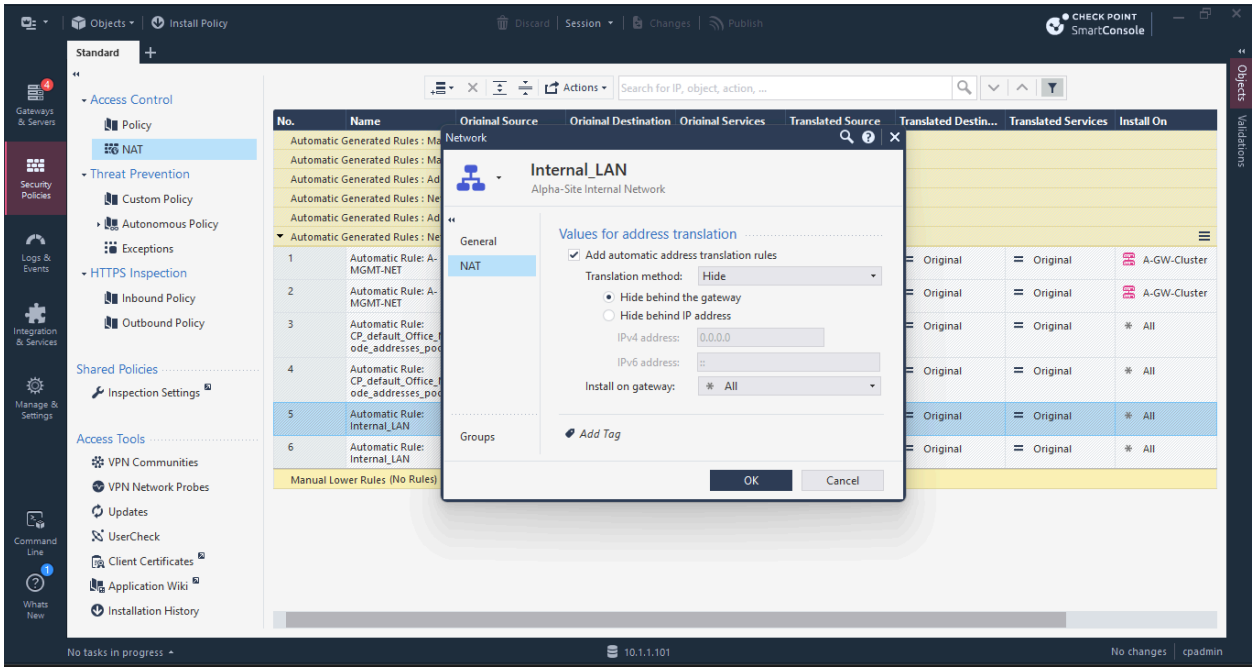
Task 3: Organize the rulebase with sections

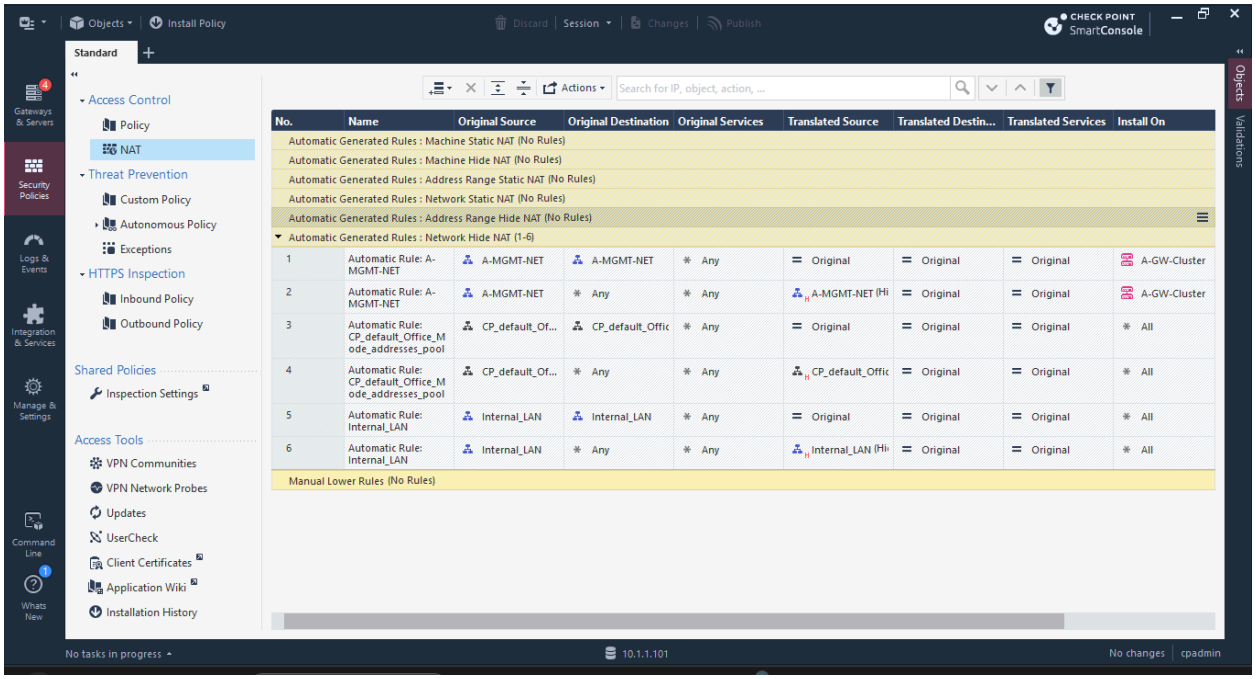




Policy sections are used only to organize and group rules for easier administration. They **do not change the rule enforcement order**, as the Security Gateway still evaluates rules from top to bottom according to their position in the rulebase. Policy sections themselves **are not sent to the Security Gateway**; only the access control rules contained within the sections are compiled and installed as part of the security policy.

Task 4: Configure Hide NAT



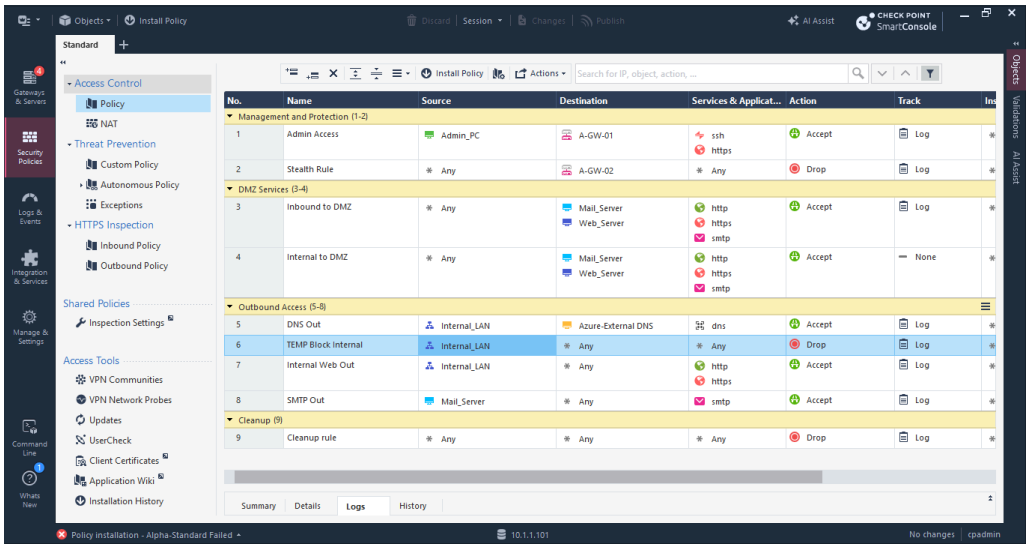


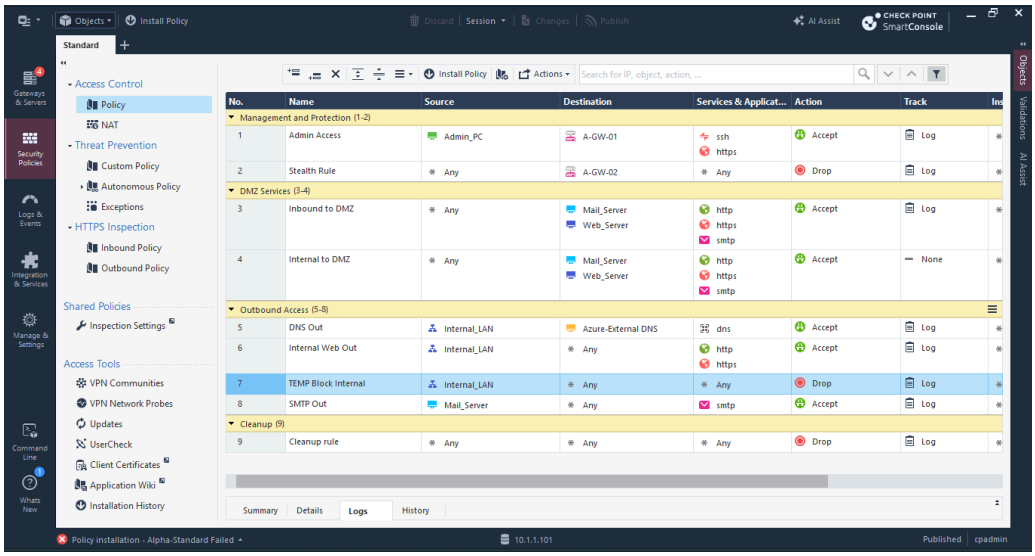
This is an automatic NAT rule because it was created by enabling the “add automatic address translation rule” on the Internal_LAN network object. Checkpoint automatically creates corresponding NAT rules based on the configuration in the object NAT properties.

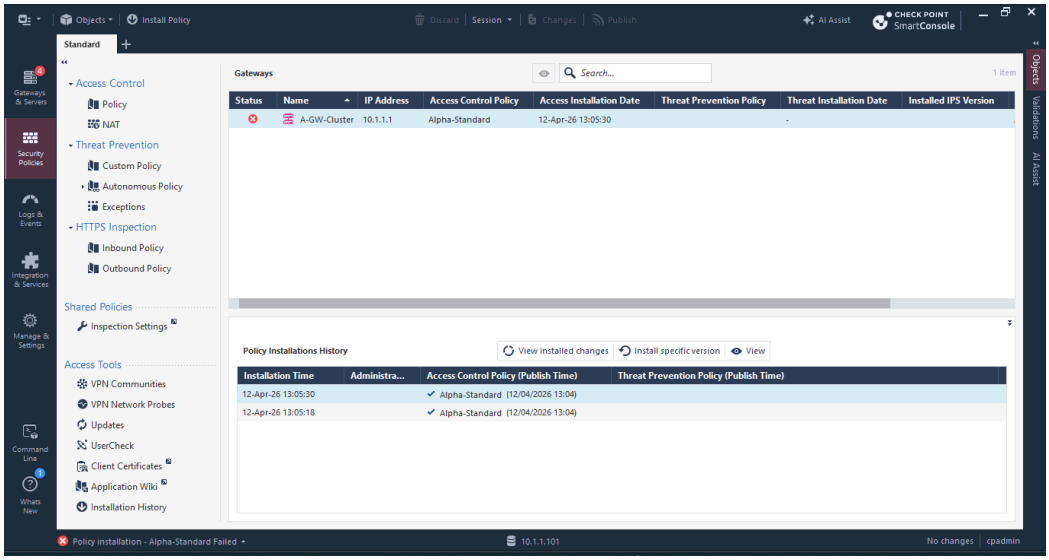
Many-to-one translations means that many internal hosts within the Internal_LAN network share a single external IP address when accessing external networks. The translation method selected, which is “Hide NAT behind the gateway” , is a many-to-one translation.

The gateway keeps track of each connection by translating the source port numbers using a process called **Port Address Translation (PAT)** ensuring that returning traffic to the internal hosts using the same public IP address are correctly sent back to the internal device.

Task 5: Rule Order experiment and installation history







Specific rules should be placed above general rules because the firewall checks rules from top to bottom and stops when it finds the first match. If a general rule is above a specific rule, the traffic may match the general rule first and the specific rule will never be used. For example, when **TEMP Block Internal** was placed above **Internal Web Out**, the traffic was blocked; when it was moved below **Internal Web Out**, the traffic was allowed.

PART B

Section B1 Answers

Q1. Where is a Security Policy created, stored and enforced?

Ans: B. Created in SmartConsole, stored on the Security Management Server, enforced by Security Gateways

Q2. Which control model does the Check Point Access Control firewall policy use?

Ans: A. Positive Control Model (whitelist)

Q3. Where should the Stealth rule be placed?

Ans: C. As early as possible, typically immediately after the Management rules

Q4. Which statement about rulebase sections is correct?

Ans: D. Sections are visual divisions only and do not affect enforcement order

Q5. What type of translation does Hide NAT perform?

Ans: B. Many-to-one

Section B2: Short answers

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Check Point security Gateway places implicit rules at the first, last or before last in the firewall rulebase but not visible in the rulebase. They apply to each layer of the Access Control Policy and are created in the global properties of the security management server. However, explicit rules are created by the administrator and configured to allow or block traffic based on specific criteria and are visible in the rule base. The Implicit Cleanup Action is a security gateway action sometimes called Implied Cleanup Rule whose options are Drop or Accept.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

Rule order is a critical aspect of an effective rulebase. Specific rules are placed before General rules to prevent general rules from being applied before specific rules. For example, when "TEMP Block Internal" was placed before "Internal Web Out", it got applied before "Internal Web Out" and the traffic was blocked.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

Saving stores the changes in the current administrator session only by creating a draft of the edited policy on the security management server. Publishing changes commits the changes to the Management Server database. During policy installation, only published changes are installed on the target Security Gateway, unpublished changes are not installed on the target Security Gateway.

Organizations publish changes only during a change window because publishing makes configuration changes official and available for deployment, to help with change control, testing, approvals, auditing and to reduce the risk of unexpected outages or security issues during business hours.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

Global Properties are enforced by all Security Gateways managed by the Security Management Server, while **Policy Packages** contain the specific security policies enforced on the gateways; It is a logical grouping of one or more policy types.

Global Properties applies to all relevant policies and gateways managed by the server while **Policy Packages** applies to a particular Gateway or group of Gateway.

To open Global properties in the SmartConsole, firstly go to Manage & Settings → Blades → Global Properties. While to open Policy Packages in SmartConsole, go to Security Policies, then select the required Policy Package from the package menu.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

The four predefined Security Zones are InternalZone, ExternalZone, DMZZone and WirelessZone.

The two rules for assigning gateway interfaces to zones are i) Each interface can belong to only one Security Zone at a time. ii) A zone must contain at least one gateway interface before it can be effectively used in Access Control rules.

The benefit of using zone-based rules is that it simplifies policy management instead of creating rules for individual networks or interfaces. It allows administrators to create a single rule based on zones, such as allowing traffic from Internal to External, making policies easier to manage and maintain.

Section B3: Scenario questions

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

1) Check Point evaluates the rules from top to bottom and since the rule is placed above the Stealth Rule and has the Any-Any-Any Accept rule it would match all traffic and bypasses the Stealth Rule. As a result, traffic directed at the Security Gateway itself is allowed instead of being blocked by the Stealth Rule.
2) It removes access control. The rule permits all traffic between all sources and destinations using any service, greatly increasing the risk of unauthorized access, malware propagation, network scanning, and other attacks.

To correct it:

1) Replace the rule with specific rules that allow only the required traffic.
2) Ensure the Stealth Rule remains above general access rules so that unwanted traffic to the gateway itself is blocked before other traffic is evaluated.
3) And lastly, follow the principle of least privilege, allowing only necessary sources, destinations, and services.

The management rule is the rule that is allowed to sit above the Stealth rule because the Stealth rule is recommended to drop any traffic destined for the Security Gateway that is not explicitly allowed.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT.

1. NAT helps to conserve public IP addresses by allowing multiple internal hosts to share a single public IP address.
2. It provides a level of security and privacy by hiding internal IP addresses from external networks, making the internal network structure less visible to attackers.

(b) Describe three checks you would make to find the fault, and explain why each one matters.

1) Verify the Hide NAT configuration on Internal_LAN
   Check that the **Internal_LAN network object** has Hide NAT enabled and is configured to hide behind the correct gateway IP address. It matters because an incorrect NAT configuration may prevent internal addresses from being translated, causing Internet-bound traffic to be dropped or routed incorrectly.
2) Confirm that the Access Control Policy allows outbound Internet traffic
   Review the Security Policy to ensure there is a rule allowing traffic from Internal_LAN to the Internet (for example, Source: Internal_LAN, Destination: Any, Service: HTTP/HTTPS, Action: Accept). This matters because even if NAT is working correctly, blocked traffic will never reach the Internet.
3) Check that the policy has been installed and verify logs
   Confirm the latest policy was successfully published and installed on the gateway. Then inspect SmartConsole logs for dropped connections or NAT-related errors. This matters because configuration changes do not take effect until the policy is installed, and logs can quickly identify whether traffic is being translated, allowed, or blocked.